Paleomagnetism of pre-Grenvillian mafic rocks from the northeast Grenville Province, Labrador: implications for the Grenville Track[1]

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Abstract: A paleomagnetic study of mafic plutonic rocks in the northeast Grenville Province found five magnetizations, interpreted as Grenvillian and later overprints. Components A, B, and C, with the highest unblocking temperatures and distinct regional distributions, have paleopoles that appear to lie on the Grenville Track. Component C, with pole at 348°E, 11°S (δp, δm = 2°, 4°) and mostly distributed in lower-grade mafic rocks in the Groswater Bay terrane near the Grenville Front, is interpreted as a post ca. 970 Ma Grenville overprint, predating components A and B. The latter components, nearly coextensive in distribution in the Groswater Bay and Sandwich Bay regions farther south, have respective poles at 326°E, 02°N (N = 9 sites; δp, δm = 8°, 11°) and 348°E, 13°N (N = 13 sites; δp, δm = 6°, 8°). Component B could be interpreted either as a Grenville overprint or, more likely, as a later overprint of late Neoproterozoic (615 Ma) to Cambrian age due to rifting in the Lake Melville region. If so, many other so-called "Grenville" overprints from rock units within or near the St. Lawrence rift system may have been erroneously attributed to Grenvillian thermal effects and, in fact, be younger. If component B is post-Grenville and component C, a Grenville overprint, then component A is probably latest Grenvillian or later Neoproterozoic in age. These results strongly suggest that the Grenville Track needs reinterpretation.

Résumé : Une étude paléomagnétique des roches plutoniques mafiques, dans le nord-est de la Province de Grenville, a révélé cinq aimantations, lesquelles sont interprétées comme grenvilliennes ou comme des surimpressions postérieures. Les composantes A, B et C, caractérisées par les plus hautes températures de blocage et des distributions régionales distinctes, ont leurs paléopôles qui apparaissent être liés au parcours des pôles de Grenville. La composante C, avec un pôle à 348°E, 11°S (δp, δm = 2°, 4°), qui est distribuée principalement dans les roches mafiques de faible degré métamorphique dans le terrane de Groswater Bay, près du Front de Grenville, est interprétée comme étant une surimpression de Grenville, postérieure à environ 970 Ma, et prédatant les composantes A et B. Ces deux dernières, réparties sensiblement sur la même étendue dans les régions de Groswater Bay et de Sandwich Bay plus au sud, ont leurs pôles respectifs à 326°E, 02°N (N = 9 sites; δp, δm = 8°, 11°) et 348°E, 13°N (N = 13 sites; δp, δm = 6°, 8°). La composante B peut être interprétée soit comme une surimpression de Grenville ou, plus vraisemblablement, comme une surimpression plus tardive, d'âge Néoprotérozoïque (615 Ma) à Cambrien, associée au développement d'un rift dans la région du lac Melville. S'il en est ainsi, de nombreuses autres surimpressions dites de « Grenville » des unités lithologiques, soit à l'intérieur ou aux environs du système du rift du Saint-Laurent, ont pu être faussement attribuées à des effets thermiques grenvilliens et, dans les faits, elles seraient plus jeunes. Si la composante B est post-grenvillienne et que la composante C est une surimpression de Grenville, alors la composante A date probablement de la fin du Grenville, ou du Néoprotérozoïque plus tardif. Ces résultats suggèrent fortement que la courbe du parcours des pôles de Grenville doit être révisée.
[Traduit par la rédaction]

Introduction

Reported here are paleomagnetic results from mafic plutonic rocks in the northeast Grenville Province. The study originated as a paleomagnetic investigation of the Mesoproterozoic Michael gabbro, conducted in conjunction with petrological studies carried out by R.F. Emslie (Emslie et al.[3]). Prior to fieldwork, it was understood that all of the mafic plutonic rocks identified during previous mapping belonged to the Michael gabbro suite and that differences in texture and mineral assemblage could be attributed to either prograde or retrograde metamorphic modifications. During sample collection, however, it became evident that more than one suite was present, introducing complications with respect to interpretation of both the paleomagnetic and petrological information obtained. However, as the paleomagnetic data presented can be interpreted as entirely due to Grenvillian

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[3] R.F. Emslie, M.A. Hamilton, and C.F. Gower. The Michael Gabbro and other Mesoproterozoic lithospheric probes in southern and central Labrador. Submitted for publication.

or later overprints, correct classification of a sample to a particular (pre-Grenvillian) suite is a secondary consideration.

Previous paleomagnetic work on Proterozoic rocks of the Grenville Province has indicated that there are two major magnetizations, called "A" and "B" (not generally the same as the A and B components of this study). Magnetization "A" has been related to regional metamorphism at about 980 Ma, and "B" to later uplift and cooling at 900 Ma and later (Berger et al. 1979). Our data, taken in conjunction with other paleomagnetic studies in the region (Murthy et al. 1989*b*, 1992), suggest that "B" may comprise several overprints, indicating that reinterpretation of the Grenville Track is in order.

Geological background

The area of study is situated in the northern part of the Grenville Province in eastern Labrador. Most of the samples were collected from the Groswater Bay terrane, but a few localities were also sampled in the adjacent Makkovik Province to the north and in the Lake Melville terrane to the south (Fig. 1).[4]

Mafic rocks are extensive in all three regions. Two suites of contrasting age but overlapping composition and mineral assemblage are now unequivocally established, but additional groups could be present.[4]

The type example for the established older group of mafic plutonic rocks is the Adlavik Intrusive Suite (Gower 1981) in the Makkovik Province (north of the area shown in Fig. 1), with an emplacement age of 1649 ± 1 Ma (Kerr et al. 1992). A mafic intrusion at Grady Island (Fig. 1), having an emplacement age of 1645 ± 3 Ma (Kamo et al. 1995), is part of the same event, as is the White Bear Arm Complex (>1641 Ma, Kamo et al. 1995) (south of the area shown in Fig. 1). All three bodies are layered intrusions showing a wide range of rock types and textures. Although gabbronoritic rocks are most common, compositions range from ultramafic cumulates to felsic differentiates. Many are multiple intrusions, crosscut by compositionally similar mafic dykes or intrusive mafic–felsic breccias, both best interpreted as feeders to younger layered units in the same intrusion. Primary amphibole and biotite are characteristic of lower grade rocks belonging to the suite. The term "Adlavik-type intrusion" is used subsequently to refer to these mid-Labradorian intrusions in eastern Labrador.

By contrast, the Michael gabbro (Fahrig and Larochelle 1972) has an emplacement age of 1426 ± 6 Ma (Schärer et al. 1986). Except for very diffuse compositional heterogeneities, lower grade Michael gabbro is a homogeneous, tholeiitic olivine gabbro. In contrast to the Adlavik-type intrusions, it lacks layering and generally lacks primary orthopyroxene and amphibole.

From the above descriptions, there would seem to be sufficient criteria to assign any particular outcrop to one or the other suite. Problems arise, however, because of a metamorphic gradation across the Groswater Bay terrane from upper greenschist facies in the north to high-pressure granulite facies in the south, interpreted to be related to Grenvillian thrusting (Gower and Erdmer 1988). As a result, primary olivine–clinopyroxene–plagioclase–biotite–opaque mineral assemblages (typical of the Michael gabbro) are replaced, first by garnet- and orthopyroxene-bearing coronitic-textured metagabbros and ultimately by two-pyroxene ± garnet granulites. It is to these assemblages that the Adlavik-type intrusions also converge. Retrograde hydrous metamorphism, generally producing garnet- or pyroxene-bearing amphibolite, introduces further complications. Chemical "fingerprinting" using a Zr/Y versus SiO_2 plot assists in distinguishing between the suites, and is discussed elsewhere.[4]

Paleomagnetic methods

At each of 84 sampling sites, generally five (2.5 cm diameter) cores were drilled, and then oriented in place using sun and magnetic compasses. Approximately 1200 cylindrical specimens (2.2 cm height) were prepared from 431 core samples by Lakefield Research of Canada, Ltd. After sample preparation, natural remanent magnetizations (NRM's) were measured from at least one specimen of every sample, either with Schonstedt DSM or Geofyzika JR-4 spinner magnetometers. Analysis of all NRM's to determine magnetic components present involved alternating-field (AF) and (or) thermal demagnetization, combined with use of vector diagrams and the LINEFIND program (Kent et al. 1983). Demagnetization was halted when magnetizations exhibited unstable behaviour or became too weak to measure. Later experiments involving thermal demagnetization and measurement of samples were conducted in a magnetically shielded room with ambient field <3000 γ. Experimental techniques are explained by Park and Jefferson (1991).

Results and analysis

NRM directions appear to be mainly concentrated in a southeast–northwest-trending streak that passes through the present Earth's field (PEF) direction in the region (Fig. 2). AF and (or) thermal treatment revealed that the NRM of individual specimens is composed of, at most, three of five identified magnetic components, two of which, D and E, are steeply directed near the PEF direction and its reverse (Fig. 3). The major components A, B, and C form two directional groupings, roughly antiparallel to one another in the northwest and southeast quadrants (Fig. 3). Taken over all sites, the unblocking temperature (T_{UB}) and coercive force spectra of all components overlap to a high degree, but in individual specimens the spectra commonly have well-defined ranges. The often discrete T_{UB} ranges of A and C are mainly suggestive of partial thermoremanent magnetizations (pTRM's). Site mean directions of all components, including the treatment ranges over which they were recovered, may be found in the Depository of Unpublished Data;[4] component means are given in Table 1.

Components B and C: Components A, B, and C have the highest T_{UB} spectra, with nearly identical ranges; the spectra of B are generally more distributed versus those of A and C. Component A generally has the lowest T_{UB}'s (Figs. 4, 5), except at site 43, where B is the least stable (Fig. 6). This is the only site where B and C occur together. At this site and

[4] More information on sites, including precise coordinates, rock types, paleomagnetic components, and identification of the mafic suites, may be purchased from the Depository of Unpublished Data, Document Delivery, CISTI, National Research Council of Canada, Ottawa, ON K1A 0S2, Canada.

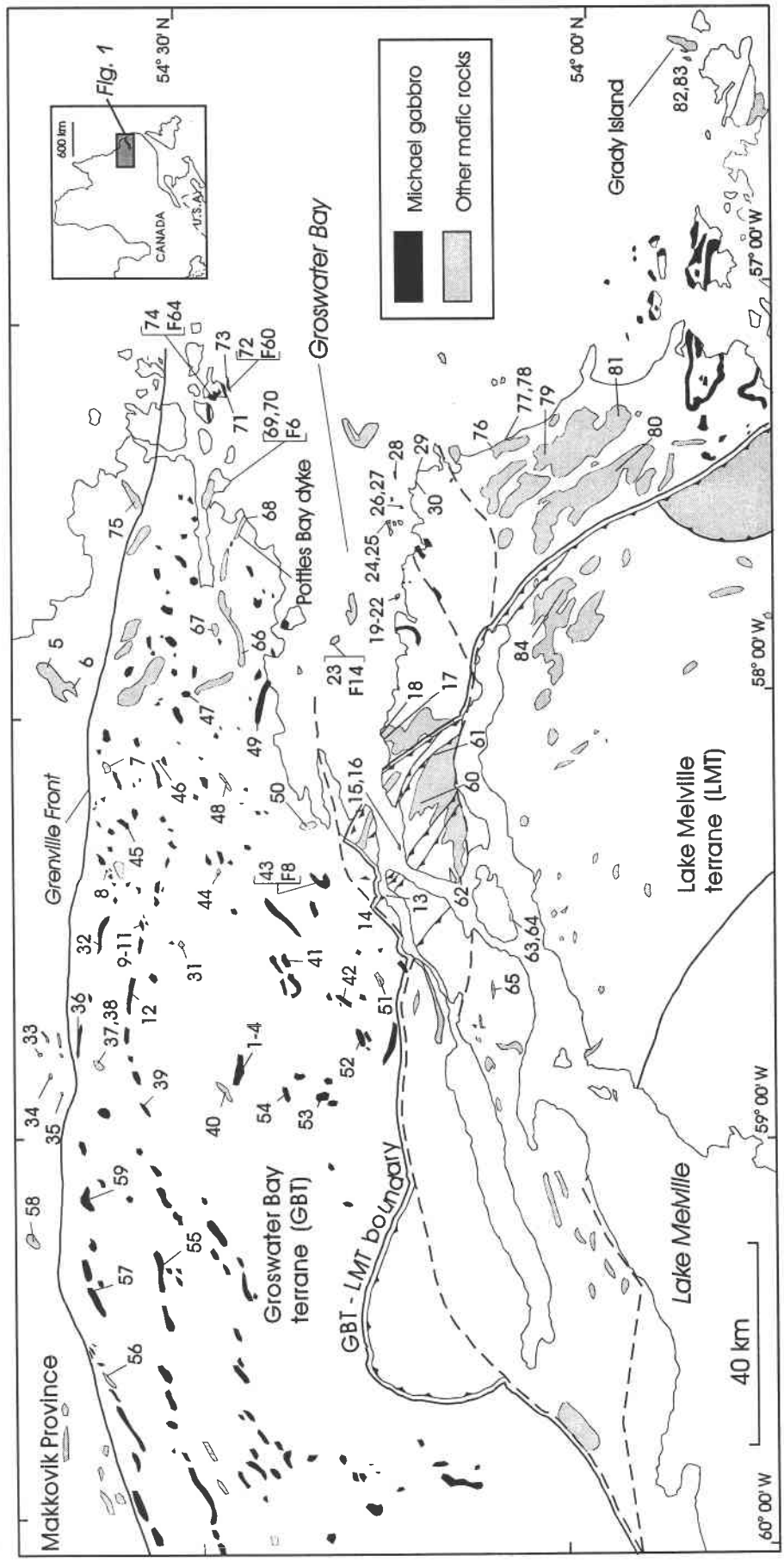


Fig. 1. Geological map of part of the northeastern Grenville Province showing paleomagnetic site locations. The "Other mafic rocks" sampled include mainly Adlavik-type intrusions. Numbers preceded by F represent sites of Fahrig and Larochelle (1972) from the same sampled intrusions. Broken lines indicate normal faults; lines with triangular symbols, thrust faults.

Table 1. Magnetic component direction means and poles.

Co	Unit wt.	N	D (°)	I (°)	R	k	α_{95} (°)	Pole °N	Pole °E	δp (°)	δm (°)	λ (°N)
A	I, S	9	331	−49	8.78	36	9	−02	326	8	11	30
B	I	11	120	+54	10.82	54	6	13	348	6	9	35
B	S	13	121	+54	12.78	54	6	13	348	6	8	35
C	I	33	133	+26	32.17	39	4	−11	348	2	4	13
C	S	36	133	+26	35.13	40	4	−11	348	2	4	14
D	I	30	351	+80	29.55	65	3	73	292	6	6	71
D	S	35	352	+80	34.51	69	3	73	293	5	6	71
E	I, S	5	352	−76	4.87	30	14	29	305	24	26	64

Notes: Co, component; Unit wt., (unit weight) (I, intrusions; S, sites.); N, number of intrusions or sites; D and I, declination and inclination of direction; R, vector resultant; k, Fisher's (1953) index of precision; α_{95}, radius of the circle of confidence about the direction with $P = 0.05$ (Fisher 1953); δp and δm, semiaxes of the error ellipse of 95% confidence about the mean pole; λ, paleolatitude at the mean site locality (54.5°N, 058.5°W).

Fig. 2. Stereographic projection of natural remanent magnetization directions from Michael gabbro specimens. ×, downward-directed magnetizations; □, upward-directed magnetizations. PEF, present Earth's field direction in the region.

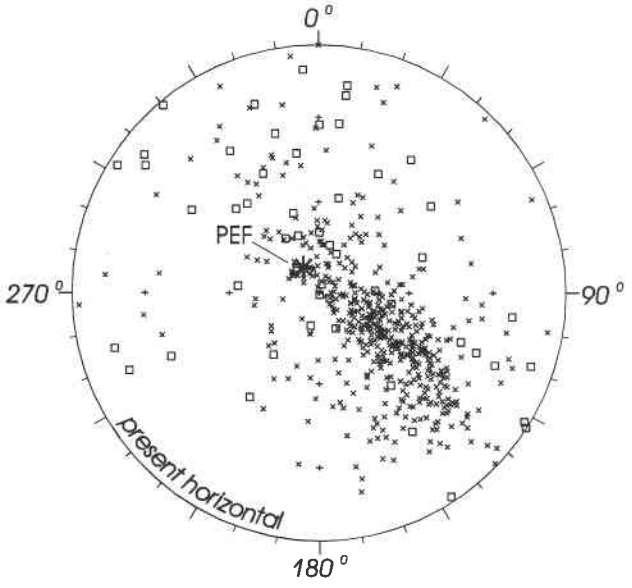
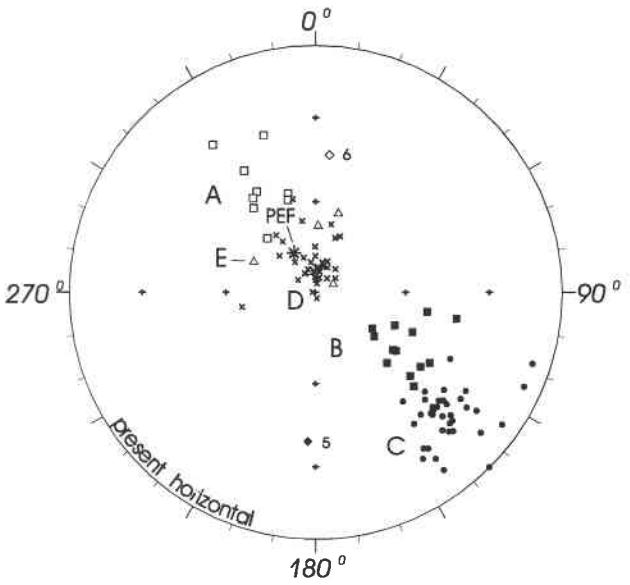


Fig. 3. Stereographic projection of site mean component directions. Solid symbols denote downward directions, and open symbols upward directions. Sites 5 and 6 are possibly Makkovikian or Labradorian in age. See also Fig. 2.



at one locality, where a dyke (site 63) containing C intrudes a layered mafic intrusion (site 64) containing B, component C has higher stability. Components B and C also have somewhat similar directions in the southeast quadrant, with B having steeper inclinations (Fig. 3). But only at site 43 and at sites 63 and 64 can they be unequivocally distinguished. At other sites, they have been identified on the basis of location: all steeply inclined components called B are in the region of Groswater Bay – Lake Melville, whereas those labelled C lie mainly to the north and on the periphery of this region (Fig. 7).

Component A: Component A is nearly coextensive in area with component B (Fig. 7). Although they are distinctly separated in individual specimens (Figs. 4, 6) and appear not to be antiparallel (A has a more northerly declination; Fig. 3), it is possible that they are identical. Statistical tests were applied to the A and B populations to test for this possibility. A ratio of their precisions (k_B/k_A = 54/36, see Table 1) gives a value of 1.50, less than the value 2.19 from F ratio tables, suggesting that there are no grounds for rejecting the null hypothesis that the two populations have equal values of k (Watson 1956). However, a test comparing the mean directions yields an F ratio of 12.58 compared with the tabled value of 3.25 ($F_{0.95}$ with 2 and 36 degrees of freedom, see Table 1), suggesting that the populations are not identical (Watson 1956).

Components D and E: Component D is prevalent over the whole sampling region, but component E is found only sporadically. Both are generally less stable than A, B, and C (Figs. 5, 8), are considered to be recent overprints, and are not discussed further.

It is difficult to relate the magnetic components to specific

Fig. 4. Vector diagram depicting the thermal demagnetization of a specimen from site 26. Solid circles indicate in situ vectors on the $X-Y$ horizontal plane, and open circles, in situ vectors on the UP$-Y$ vertical plane. Positions of removed components (letters) are approximately shown. Intensities in mA/m.

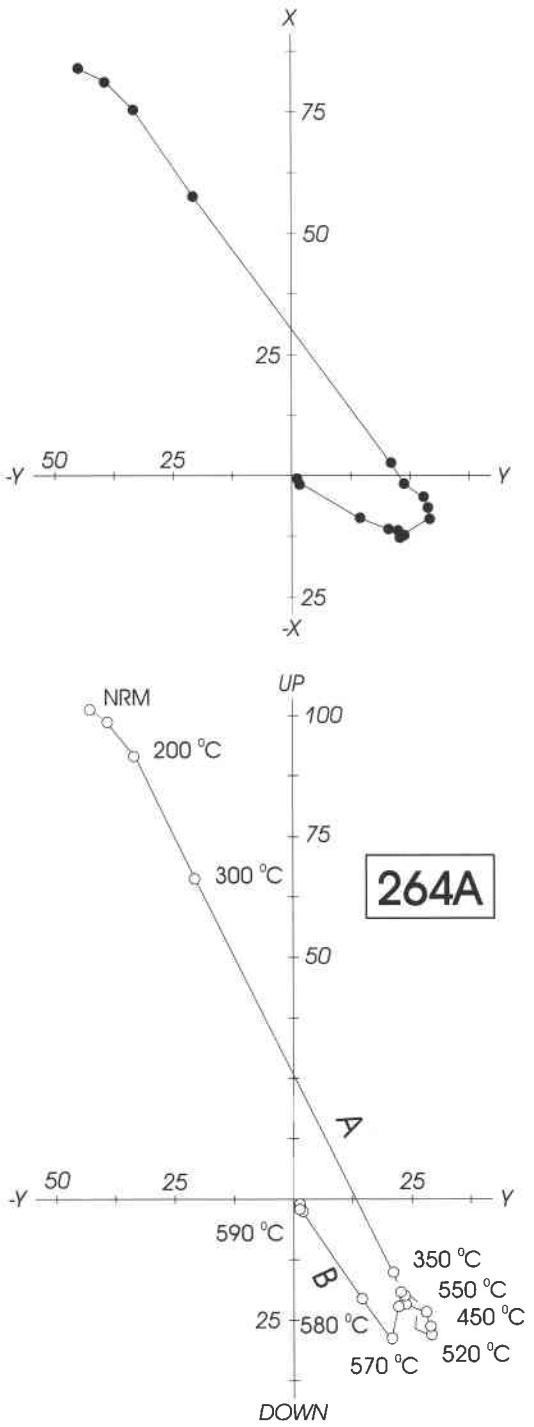
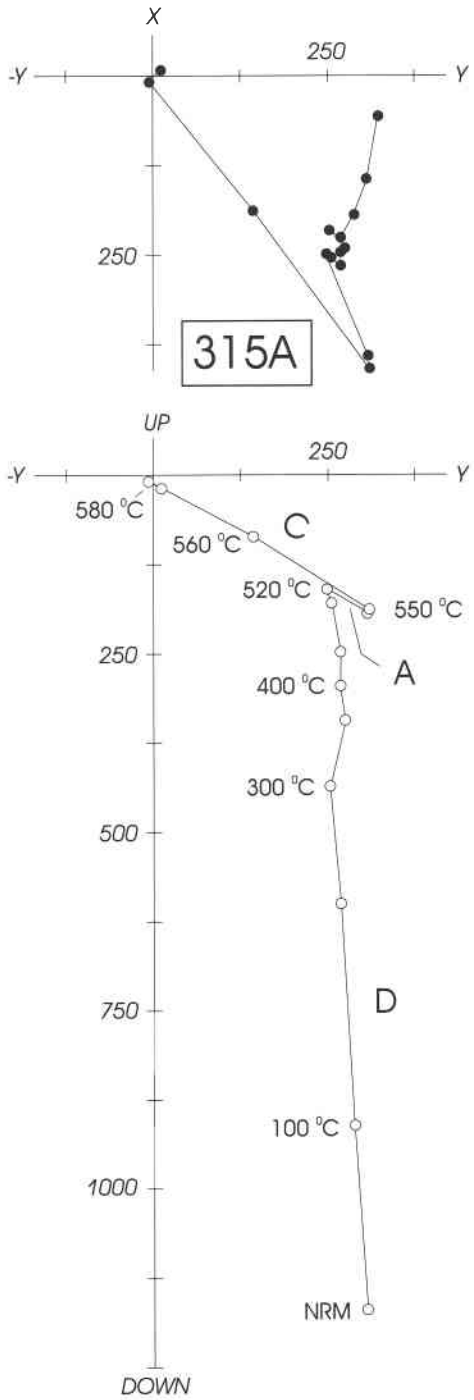


Fig. 5. Vector diagram depicting the thermal demagnetization of a specimen from site 31. Other explanations as in Fig. 4.



magnetic mineral phases because of the wide overlap of the coercivities and of the T_{UB} spectra of components, and the lack of correspondence between revealed magnetizations and identified magnetic minerals. The principal magnetic minerals, magnetite and hematite, occur in several different phases and generations within the mafic rocks, as summarized elsewhere.[4] Revealed T_{UB}'s indicate that magnetite and hematite harbour part of components A, B, and C. Sulphides are common, but it is uncertain whether the commonly identified magnetic sulphide pyrrhotite contains any of the magnetic components. Its characteristic T_{UB}'s and coercive forces relate more readily to D and E.

Discussion

Magnetic components

Component directions are significantly different from one another (Table 1) and appear to have regional significance (Fig. 7). Component C, the characteristic magnetization of most of the mafic rocks of this study, is concentrated within the northern Groswater Bay terrane, the region where most of the previous sampling of the Michael gabbro (Fahrig and Larochelle 1972; Murthy and Deutsch 1972) was carried out. In the Smokey Islands of Groswater Bay, the present results (sites 71–74; all Michael gabbro) corroborate those of Fahrig and Larochelle (1972) but not those of Murthy and Deutsch (1972) (Table 2), indicating the latter results to be in error. Fahrig and Larochelle (1972) earlier characterized the magnetization as primary; here, it is ascribed to a secondary magnetization. Three sites recording C come from the Makkovik Province, just north of the Grenville Front (sites 33–35; interpreted to be Adlavik-type intrusions).

Some anomalous C directions were excluded from the results: that is, directions with inclinations less than 18° (sites 19, 23, 31, 35, 37, 60, 77; all Adlavik-type intrusions), whose sites, except for 60, appear to be aligned over a distance of 150 km (Fig. 7); and two directions (sites 5, 6) from rocks of likely Makkovikian, or perhaps Labradorian, age in the Makkovik Province (Figs. 1, 3). Derived poles from sites 5 and 6 lie near poles considered to be 1850–1800 Ma in age.

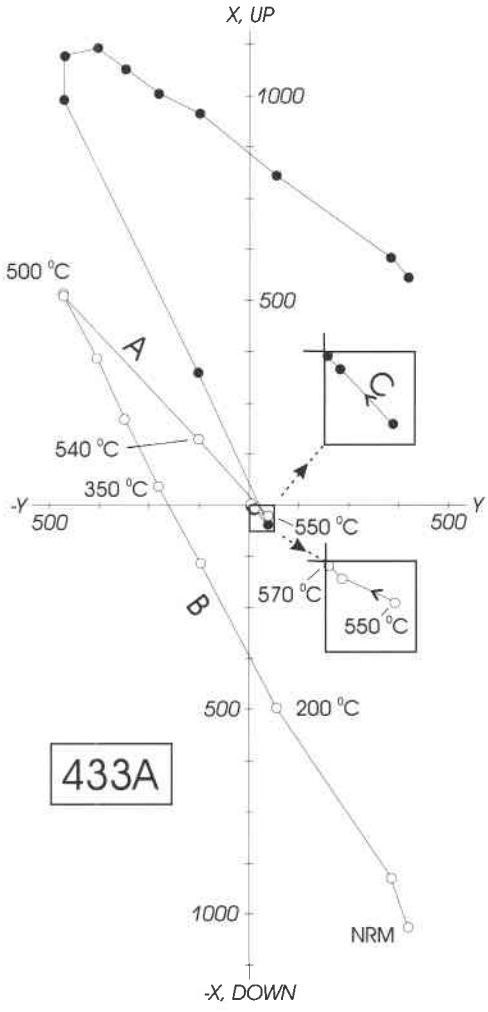
In general, components A and B occur in mafic rocks to the south of those carrying C, and, including results from earlier studies (Murthy et al. 1989*b*, 1992), appear to be nearly coextensive in area (Fig. 7). Components A and B only rarely occur with C (e.g., Figs. 5, 6).

Component A is identical to the characteristic magnetization analyzed by Murthy et al. (1989*b*) from "pre-Grenvillian mafic intrusions" in the Sandwich Bay region, including the 1645 ± 3 Ma Grady Island intrusion and 1426 ± 6 Ma Michael gabbro locality (Table 2). A similar component was revealed in the White Bear Arm Complex (Hale 1988). In all three cases, the magnetizations have similar directions to Grenville "B" components from the central Grenville Province, suggesting that in their respective areas, the Michael gabbro and Adlavik-type intrusions have been overprinted.

Magnetic poles and interpretation

Paleopoles of A, B, and C are plotted in Fig. 9 along with other paleopoles from Laurentia thought to be of late Mesoproterozoic to Neoproterozoic age. Components A, B, and C appear to lie on the Grenville or a later apparent polar wander path. The thermochron model of Grenville poles (McWilliams and Dunlop 1978), which relates poles to the cooling ages on thermochron maps, was used in the past to establish a chronology of Grenville poles (see McWilliams and Dunlop 1978 and references therein), later refined by Berger et al. (1979). Thus, poles were related to the track by interpretation of associated components as pTRM's. Concerning the thermochron model, Roy and Robertson (1979) pointed out problems with inherent assumptions of cooling as single stage and of magnetizations as thermoremanent in origin. In addition, the standard Grenville Track of early Neoproterozoic poles from the Grenville Province has a large bulge in the vicinity of Grenville "B" poles, dated at about 900 Ma

Fig. 6. Vector diagram depicting the thermal demagnetization of a specimen from site 43. Other explanations as in Fig. 4.



(Fig. 9*a*). Combining the reservations of Roy and Robertson (1979) with the fact that A, B, and C poles are all distributed around the bulge suggests that the Grenville Track requires reinterpretation.

An earlier interpretation suggested that poles MG (C of this study), SC_2, and MA_{NW} were displaced from similar-aged poles (ca. 1450–1300 Ma) (mean of poles, EP; Fig. 9*b*) from elsewhere in North America due to rotation of a Makkovik block (Fahrig and Larochelle 1972; Roy and Fahrig 1973; Fahrig et al. 1974). This interpretation was questioned by Roy and Robertson (1979), who suggested that the poles might instead be derived from secondary remanences acquired at ca. 950 Ma. The fact that the C component is found in both Adlavik-type intrusions and Michael gabbro, both subjected to high grades of metamorphism, confirms its secondary nature.

Poles MG and C appear to relate to the southward-trending Grenville Track. That they are related to this track is consistent with poles MD_A (Mealy dykes) and MA_{NW} (Mealy Mountains anorthosite) from the structurally higher Mealy Mountains terrane, less affected by Grenvillian metamorphism, which are interpreted to lie on the earlier Keweenawan

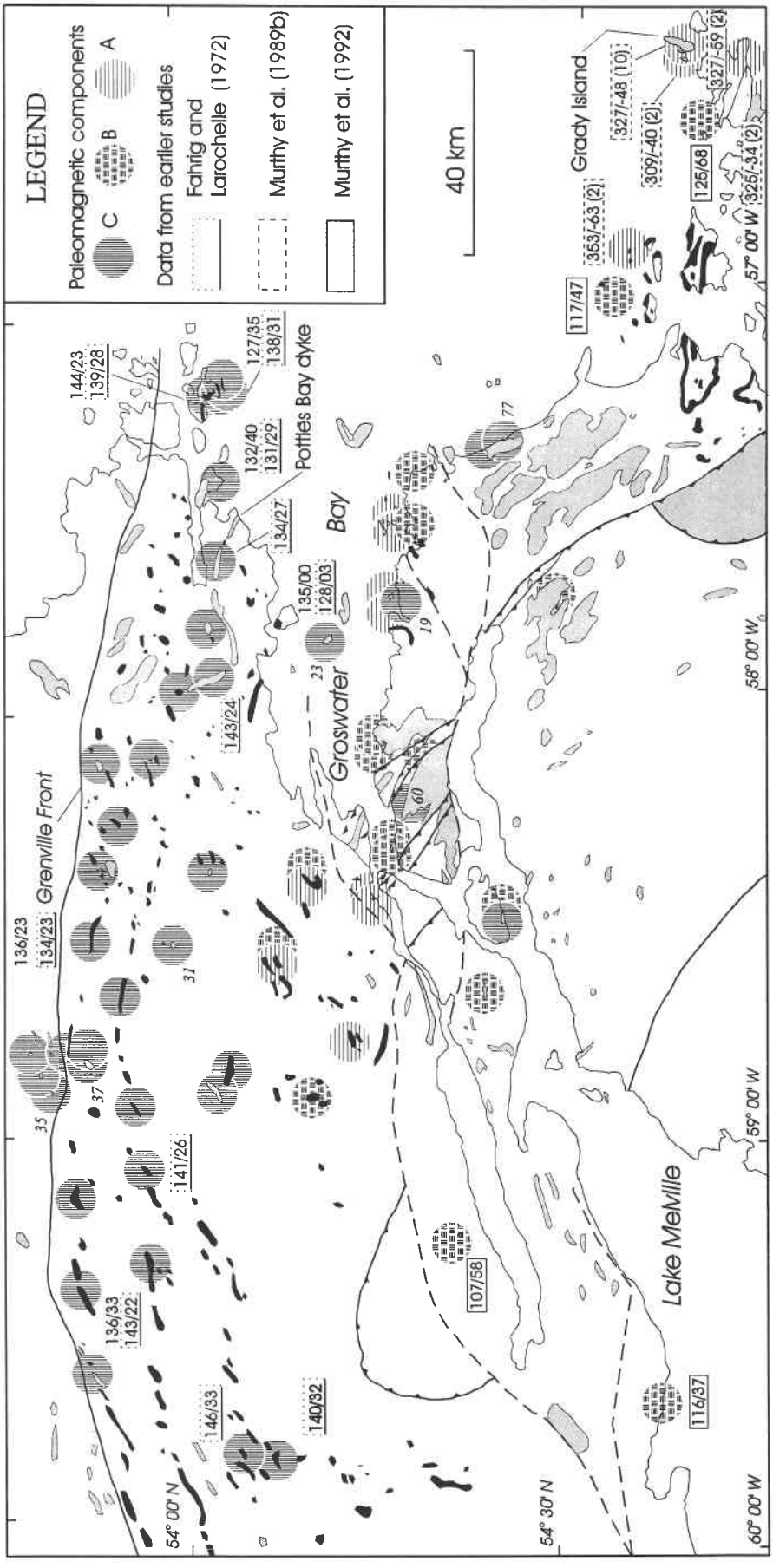


Fig. 7. Map showing distribution of magnetic components at sites indicated in Fig. 1. Coverage is extended using component directions (declination/inclination enclosed by rectangles) from earlier studies. Numbers in parentheses indicate the number of sites represented by the data. Data from this study (not in rectangles) from common sampled intrusions are shown for comparison. Where site numbers are shown, the inclination of the C component is $<18°$. See Fig. 1.

Fig. 8. Vector diagram depicting the thermal demagnetization of specimens from sites 23 (*a*), 65 (*b*), and 33 (*c*). Other explanations as in Fig. 4.

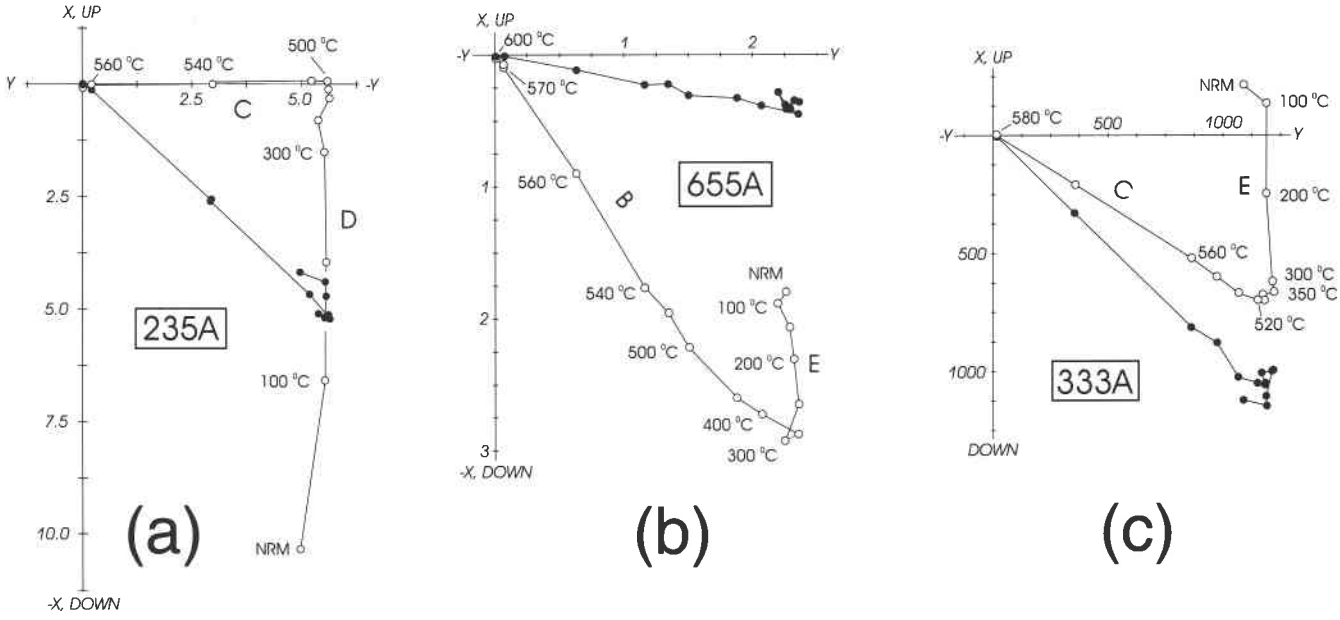


Table 2. Comparison of paleopoles from this and previous studies in the region.

Co	Unit	*N*	*D* (°)	*I* (°)	*k*	α_{95} (°)	Pole °N	Pole °E	δp (°)	δm (°)	Ref.[a]
A	Pre-Grenvillian mafic intrusive	9	331	−49	37	9	−02	326	8	11	1
A	Pre-Grenvillian mafic intrusive	20	322	−49	24	7	00	335	6	9	2
	Sandwich Bay dykes	11	326	−27	91	5	−16	338	3	5	3
B	Pre-Grenvillian mafic intrusive	13	121	+54	54	6	13	348	6	8	1
	Long Range dykes	3[b]	125	+56	48	18	11	344			4
	Double Mer Fm.	5	111	+50	25	16	13	359			4
C	Pre-Grenvillian mafic intrusive	36	133	+26	40	4	−11	348	2	4	1
C	Pre-Grenvillian mafic intrusive	22	137	+30[c]	152	3	−11	343	2	4	5
C	Michael gabbro	12	242	+22	15	12	−06	243	6	12	6

Notes: *N*, number of sites. See Table 1 for other explanations.

[a]Reference: 1, this study; 2, Murthy et al. (1989*b*); 3, Murthy et al. (1989*a*); 4, Murthy et al. (1992); 5, Fahrig and Larochelle (1972); 6, Murthy and Deutsch (1972).

[b]Dyke unit weight; 12 sites.

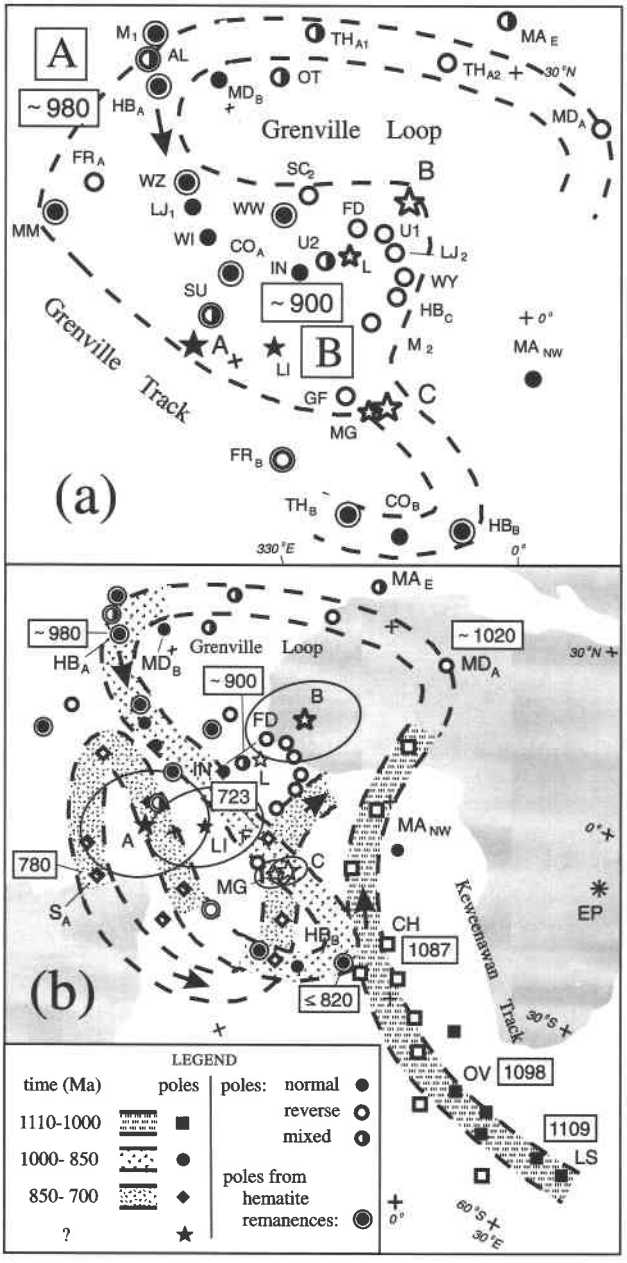
[c]Including site 14, which was not included in the group mean of Fahrig and Larochelle (1972).

Track (Fig. 9*b*) (Park and Emslie 1983). At the older end of the Grenville Track, a later overprint pole MD_B from the Mealy dykes supports this interpretation (Park and Emslie 1983; Reynolds 1989), which is based on arguments that associated remanences are largely thermoremanent in nature (see also Park and Emslie 1983).

But how do poles A and B, which appear also to be on the Grenville Track, relate to C? Pole A plots with the equivalent pole LI (Murthy et al. 1989*b*), either on the Grenville Track or later Neoproterozoic path (Fig. 9*b*). Pole LI is unlikely to be older than ca. 970 Ma, based on a concordant titanite age (Kamo et al. 1995) from Grady Island. Murthy et al. (1989*b*) interpreted a Grenville age of ca. 900 Ma. If A and C are pTRM's, A with lower T_{UB}'s in generally higher grade metamorphic terranes likely postdates C. Thus, A could relate to the later Neoproterozoic path (Park and Aitken 1986; Park 1994). A U−Pb zircon age of 615 Ma from the Long Range dykes (Kamo et al. 1989; Kamo and Gower 1994), which do not evidence A, would provide a minimum age.

Pole B appears to lie on an older part of the Grenville Track than C. At site 29, where B is the characteristic component, ages of 968^{+7}_{-8} Ma (titanite) and 930−920 Ma (rutile) were obtained from granodiorite and metagabbro, respectively (Schärer et al. 1986). Blocking temperatures of 600 and 400°C from titanite and rutile, respectively, would thus tend to constrain the age of B, which has unblocking temperatures from 550 to 580°C. However, another explanation is that B represents a much later remanence developed as a chemical remanent magnetization (CRM) during incipient rifting in late Neoproterozoic times. A CRM is more likely to

Fig. 9. Apparent polar wander path (APW) of Laurentia for the late Mesoproterozoic to Neoproterozoic period. (*a*) Grenville poles. (*b*) Interpreted Grenville Track (this paper), Keweenawan Track (Ernst and Buchan 1993), and later Neoproterozoic path (Park and Aitken 1986; Park and Jefferson 1991; Park 1994) (shaded portions of APW path). Poles of this study: A, B, and C. Other poles of Labrador: L, Long Range dykes (Murthy et al. 1992); LI, Labrador intrusions (Murthy et al. 1989*b*); MA_E and MA_{NW}, Mealy Mountains anorthosite (Fahrig et al. 1974); MD_A and MD_B, Mealy dykes (Park and Emslie 1983); MG, Michael gabbro (Fahrig and Larochelle 1972). Other poles of the Grenville Province, which are referenced by Dunlop and Stirling (1985) and (or) Park (1994): AL, Allard Lake anorthosite; CO_A and CO_B, Cordova gabbro; FD, Frontenac dykes; FR_A and FR_B, French River anorthosite; GF, Grenville Front anorthosite; HB_A, HB_B, and HB_C, Haliburton intrusions; IN, Indian Head anorthosite; LJ_1 and LJ_2, Lac St-Jean anorthosite; M_1 and M_2, Morin anorthosite; MM, Magnetawan metasediments; OT, Ottawa intrusions; SC_2, Seal–Croteau igneous rocks; SU, St-Urbain anorthosite; TH_{A1}, TH_{A2} and TH_B, Thanet gabbro complex; U1 and U2, Umfraville gabbro; WI, Wilberforce pyroxinite; WW, WY, and WZ, Whitestone anorthosite. Key dated poles from Laurentia: CH, Copper Harbor conglomerate; F, Franklin diabase; LS, Logan sills; OV, Ostler volcanics; S_A, diabase in the Mackenzie Mountains. EP is the mean of Elsonian poles (ca. 1450–1300 Ma). Poles of the Mesoproterozoic period are referenced in Ernst and Buchan (1993, and references therein). Errors about poles are δp and δm (see Tables 1 and 2).



have the distributed T_{UB} spectra evident in B than the more discrete T_{UB} spectra of A and C, characteristic of a pTRM.

This alternative interpretation involves the 615 Ma Long Range dykes, which crop out around and within the Sandwich Bay Graben (Fig. 10). The Sandwich Bay and Lake Melville grabens document late Neoproterozoic incipient rifting that eventually led to formation of the Iapetus Ocean in the Early Cambrian. These grabens may be related to the St. Lawrence Rift System (Gower et al. 1986) from which similar ages ranging to Early Cambrian have been reported, and which is near or encompasses all localities from which poles of the Grenville Track have been derived (Fig. 10).

The tie-in with the Long Range dykes relates to the similarity of pole B with the Long Range pole L (Murthy et al. 1992). Pole L has been interpreted as either a 615 Ma primary pole (Murthy et al. 1992; Kamo and Gower 1994) or a later Cambrian pole (Meert et al. 1994). Whether the Long Range pole L is primary or secondary, its position among the 900 Ma Grenville "B" poles suggests the possibility that other poles in the grouping represent, or have been affected by, younger events, and represent chemical overprints, as suggested by previous arguments (Roy and Robertson 1979).

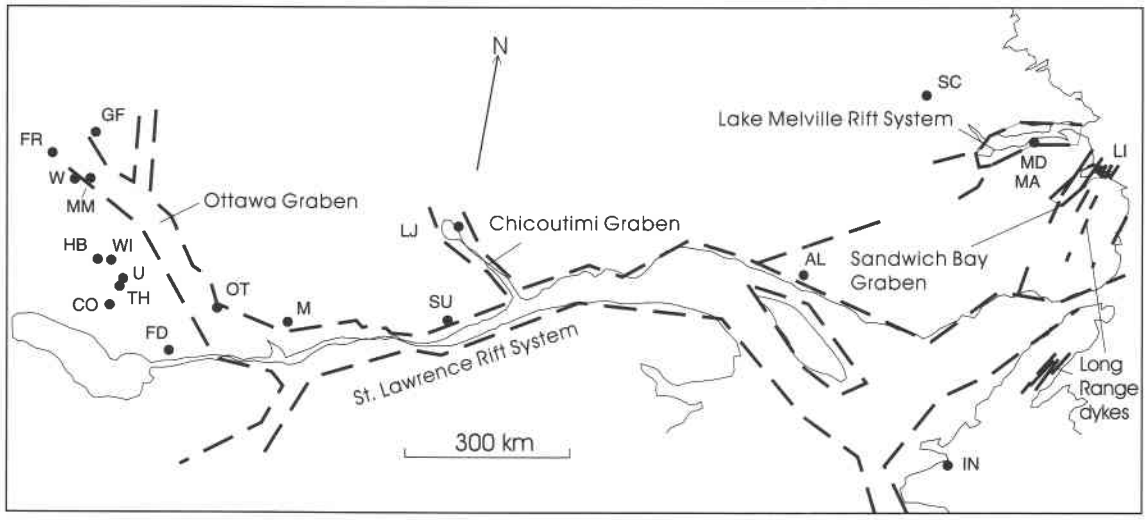
Therefore, several so-called Grenville poles could relate to the north-trending later Neoproterozoic path interpreted by Park and Aitken (1986); others may represent, or be affected by, younger rifting events. In view of these possibilities, the Grenville Track remains highly uncertain. The Grenville Track depicted in Fig. 9*b* represents one possible configuration.

Conclusions

Five magnetic components have been identified in mafic intrusive rocks in the northeast Grenville Province (Table 1). Components A, B, and C, with the highest unblocking temperatures, have distinct regional distributions.

Two mafic intrusive suites (1649–1645 Ma Adlavik Intrusive Suite plus correlatives and the 1426 Ma Michael gabbro) both show the same paleomagnetic signatures, independent of structural attitude or metamorphic grade, thus demonstrating that the magnetic components are overprints that could not have been acquired before the late Grenvillian. Component C is interpreted to be the oldest, which, from available geochronological data, was unlikely to have been acquired before 970 Ma. Component A is interpreted as a late-Grenvillian or later Neoproterozoic signature. Although component B could also be Grenvillian, our preferred interpretation is that it represents a chemical overprint associated

Fig. 10. St. Lawrence Rift System and its extension into eastern Labrador. Based on Kumarapeli and Saull (1966) and Gower et al. (1986). Approximate localities of paleomagnetic studies are plotted. Locality abbreviations as in Fig. 9.



with late Neoproterozoic or Early Cambrian rifting. No matter how the data are evaluated, it seems inescapable that the present interpretation of the Grenville Track is inadequate.

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